



Connect a Web App to Amazon Aurora

M

Mohammed Dawoud Mota

Create database [info](#)

Choose a database creation method

☒ **Full configuration**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [info](#)

☒ **Aurora (MySQL Compatible)** 

☐ **Aurora (PostgreSQL Compatible)** 

☐ **MySQL** 

☐ **PostgreSQL** 

☐ **MariaDB** 

☐ **Oracle** 

☐ **Microsoft SQL Server** 

☐ **IBM Db2** 

Engine version

Aurora MySQL 3.10.3 (compatible with MySQL 8.0.42) - default for major version 8.0

☐ **Enable RDS Extended Support** [info](#)
Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [Amazon Aurora documentation](#).

Templates

Choose a sample template to meet your use case.

☐ **Production**
Use defaults for high availability and fast, consistent performance.

☒ **Dev/Test**
This instance is intended for development use outside of a production environment.



Introducing Today's Project!

What is Amazon Aurora?

Amazon Aurora is a relational database that's built for high performance and large scale applications. It uses clusters with a writer and reader instance, which keeps your data available and gives you automatic failover and backups. It's designed for big workloads that need speed, reliability and strong uptime.

How I used Amazon Aurora in this project

I used Amazon Aurora by creating a brand-new Aurora MySQL database cluster and configuring it from scratch. I set the engine type to Aurora MySQL, chose the Dev Test template, created the cluster identifier and admin credentials, and launched the database with a writer and reader instance. I also connected the Aurora database to the EC2 instance I created so it could act as the web app server.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was seeing how Aurora automatically creates a whole database cluster with a writer and reader instance. I didn't realize it would show multiple databases in the list right away, and it was interesting to learn that Aurora uses clusters to keep everything highly available and ready for failover.



Mohammed Dawoud M...

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This project took me...

This project took me around an hour to complete.



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In the first part of my project...

Creating an Aurora Cluster

A relational database stores data in tables with rows and columns, allowing relationships between data points. It uses SQL to easily manage, query, and organize information efficiently.

Aurora is a good choice when you need a relational database to support something large-scale with peak performance and uptime.

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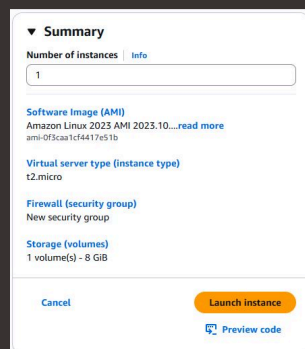
Halfway through I stopped!

I stopped creating my Aurora database because I needed to create an EC2 instance so that I could connect the web app to the database.

Features of my EC2 instance

I created a new key pair for my EC2 instance because this is what will allow me to access my instance and perform any of the needed modifications to the instance.

When I created my EC2 instance, I took particular note of the IPv4 address and the Key pair name. These are important because IPv4 tells you where the server is located, and the key pair is what allows you into the server. Without both, you can not get into the server.





Then I could finish setting up my database

The screenshot shows the 'Connectivity' settings for an AWS Aurora database instance. The page is titled 'Connectivity' with an 'Info' link. Under the 'Compute resource' section, there are two radio buttons: 'Don't connect to an EC2 compute resource' (unselected) and 'Connect to an EC2 compute resource' (selected). Below this, the 'EC2 instance' section shows a dropdown menu with the selected instance 'i-0798c582b6eae9407' and the label 'nextwork-ec2-instance-web-server'. A warning message states: 'Some VPC settings can't be changed when a compute resource is added. Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.' The 'Network type' section has two options: 'IPv4' (selected) and 'Dual-stack mode' (unselected).

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

EC2 instance [Info](#)
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-0798c582b6eae9407
nextwork-ec2-instance-web-server

Some VPC settings can't be changed when a compute resource is added
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Network type [Info](#)
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Aurora Database uses clusters because a database cluster in Aurora is a group of database copies that work together, so your data is always available. Each cluster consists of a primary instance (where all write operations occur) and multiple read replicas as back-ups. If your database's primary instance fails, one of the replicas can be promoted to primary automatically.